

ASSIGNMENT
BMIS2003 BLOCKCHAIN APPLICATION DEVELOPMENT
ACADEMIC SESSION: 202605

GROUP:

Group of 4 to 5 members

WEIGHTING TOWARDS THE ASSIGNMENT:

Total: 71%

(This assignment contributes 71% to the final coursework marks)

SUBMISSION DEADLINE:

Week 12 (Submit by 11:59pm Sunday of week 12)

LEARNING OUTCOME BEING ASSESSED:

- Produce one Blockchain-based application using a selected Blockchain platform. (CLO2)
- Cooperate to build one Blockchain-based application in a team. (CLO3)

Each team to design and develop a blockchain based application using Ethereum platform and Solidity language for **ONE (1)** of the problem statements below:

1. Decentralized Escrow and Milestone-Based Logistics Platform

Traditional supply chain logistics and trade finance heavily rely on centralized intermediaries, paper-heavy documentation, and manual compliance checks, often leading to payment delays and friction between shippers and carriers. A decentralized escrow platform leverages smart contracts to lock funds securely on-chain and distribute them automatically to service providers only when verifiable, immutable milestones are fulfilled.

Each team is to design and develop a decentralized application (dApp) using the Ethereum platform and Solidity language that enables commercial parties to create, fund, and execute milestone-based logistics agreements:

Requirements:

- **User Registration and Authentication:** Users should be able to register and log in to the platform as either a Shipper (buyer) or a Carrier (service provider).
- **Agreement Creation:** Shippers should be able to create new logistics contracts with details such as total payload value, required milestone checkpoints, and a strict delivery deadline.
- **Funding Mechanism:** Shippers must be able to contribute cryptocurrency (e.g., Ether) to lock the contract funds into the escrow smart contract upon initialization.
- **Milestone & Payout Management:** Payments must be securely held on-chain and automatically released progressively (e.g., partial payment on pickup, balance on final delivery) to the Carrier based on cryptographic verification of milestone completions.

- **Automatic Refunds / Dispute Handling:** If a Carrier fails to meet a critical milestone within the deadline, the smart contract should automatically trigger a refund mechanism back to the Shipper.
- **Transaction History:** Users should be able to view their contract participation, current escrow balances, and chronological milestone history.

For this assignment:

- You may define a token standard (e.g., ERC-20) to represent carrier reputation points or service stakes awarded upon successful milestone completions.
- You need to define the relationship between contract funds and milestone distributions (e.g., "30% released upon Milestone 1 verification").
- It is mandatory to integrate the user interface with the smart contracts. If the project is presented solely in Remix, it will be considered incomplete.

2. Decentralized Blue Carbon Registry and Offset Marketplace

The voluntary carbon market faces severe criticism due to a lack of transparency and double-counting of environmental offsets. Blue Carbon ecosystems, specifically seagrass meadows, have immense carbon sequestration capabilities. A decentralized asset exchange can bring trustless verification to this sector by minting verified carbon offsets as digital assets, allowing transparent tracking from initial ecological verification to final corporate retirement.

Each team is to design and develop a blockchain-based application using the Ethereum platform and Solidity language to manage and trade blue carbon credits transparently:

Requirements:

- **User Registration and Authentication:** Users should be able to register and log in to the platform as either an Ecological Project Developer or a Corporate Buyer.
- **Project Creation & Validation:** Project developers should be able to register carbon offset projects with details such as geographic location, ecosystem type (e.g., seagrass), target carbon sequestration capacity, and audit deadline.
- **Tokenization Mechanism (Minting):** The smart contract must mint digital assets (tokens) representing verified sequestered carbon. You need to define the relationship between carbon data and token rewards (e.g., "1 token = 1 Metric Ton of \$CO_2\$ sequestered").
- **Marketplace Trading:** Corporate buyers must be able to contribute cryptocurrency (e.g., Ether) to purchase active carbon tokens directly from project developers without intermediaries.
- **Retirement / Burning Logic:** To claim an offset, buyers must be able to permanently "retire" (burn) tokens via the smart contract, automatically updating the ledger to prevent double-counting.
- **Transaction History:** Provide a user interface to view token ownership lifecycles, historical marketplace prices, and final retirement status on a scannable table.

For this assignment:

- You may define a token standard—such as ERC-20 for fractionalized carbon credit pools or ERC-721 (NFTs) to represent unique conservation plots.
- The application must include state-handling optimization to avoid the rescan of old events on system reboot to reduce duplicate entries and sync time.

- It is mandatory to integrate the user interface with the smart contracts. If the project is presented solely in Remix, it will be considered incomplete

DELIVERABLES:

1. PROTOTYPE

Create a prototype of your application and submit it to your tutor.

2. DESIGN DOCUMENT

Create a design document of your app and submit it to your tutor.

Your design document should contain:

- Introduction
- Policy/Business Rules
- Contract Diagram
- Activity Diagram
- Use Case Diagram
- System Architecture/Design Structure
- UI Design
- Limitations/challenges faced

Note: Please use the front cover page provided for your design document.

3. PRESENTATION

Complete the assignment and present it to your tutor using presentation slides. Prepare a **20 minutes** presentation (about 5 minutes for each team member). Your presentation session will include the following:

1. Introduction
2. Design of your application
3. Live demo of your application developed
4. Code inspection by your tutor

SUBMISSION GUIDELINES

Submit the following items to your tutor before the deadline of submission: (To be compiled by the team leader and submit to your tutor)

1. Presentation slides to be used in the presentation
2. A compressed copy of your program source code (zip format)
3. Installation files (including a setup guide)
4. Design document (use the cover page template provided for this document)
5. A copy of peer evaluation form for each team member (See Appendix A. Peer evaluation is used to assess teamwork)
6. A copy of declaration form for each team (See Appendix B)
7. A copy of the task description for each team member (See Appendix C)
8. A copy of the assignment evaluation form for each team member (See Appendix D)

EVALUATION

Assessment Criteria:

Criteria	Marks
Part A (Group assessment – 70%)	
Quantity of work	20
Quality of work	20
Originality and understanding	10
Innovativeness	10
Design documentation	10
Part B (Individual assessment - 30%)	
Project involvement	10
Teamwork	10
Quality of presentation	10
Total	100

LATE SUBMISSION

No late assignments will be accepted (**get zero**). Please do not argue with your tutor if you really failed to submit your assignment on time as the consequence on late submission has been given in advance.

However, in certain circumstances, the students may be allowed to turn in the assignment late. The students must contact the tutor BEFORE the assignment is due. The tutor will evaluate whether the circumstance warrants submitting the assignment late. A late penalty will be applied. The penalty is as follows:

- Late submission within 1 - 3 days total marks to be deducted is 10 marks.
- Late submission within 4 - 7 days: total marks to be deducted is 20 marks.
- Late submission after 7 days: reject coursework and zero marks shall be awarded.

Please refer to TAR UMT guidelines on late submission of coursework for more detail.

NO-CHEATING POLICY

A reminder on the no-cheating policy: You are **NOT** to share your work with your peers, but please feel free to have a discussion with your peers. If cheating is discovered, both parties will take the equal blame (**get zero**). Please note that the assignment should be your own work, although you may incorporate ideas or techniques from books, online resources, etc. By copying materials directly from any sources of materials will lead to zero. You have been warned. Whenever you face any problems, please seek advice from your tutor.